

- (c) Rana tigrina (d) China rose
(e) None of the above / More than one of the above

B.P.S.C. CDPO 2022

Ans. (c)

Binomial nomenclature is a system of nomenclature in which each species of animal or plant receives a name of two terms of which the first identifies the genus to which it belongs and the second the species itself. Rana tigrina is an example of the Binomial scientific name of frog.

Cell

1. Plants and animal cells differ in which of the following structure?

- (a) Enzymes (b) Nuclei
(c) Cell wall (d) More than one of the above
(e) None of the above

BPSC (Tre-3) {Class 9-10}–2024

Ans. (c)

Plant and animal cells differ in several structures. Here are the few key differences :

Cell wall–Plant cell have a rigid cell wall made of cellulose, which provides structural support, whereas, animal cells do not have cell wall.

Chloroplasts–Plant cell contain Chloroplasts, which are responsible for photosynthesis whereas animal cell do not have chloroplasts.

Vacuole–Plant cells have a large central vacuole and animal cells may have small vacuoles, but they are much smaller & less prominent.

Plant and animal cells share some similarities, such as a nucleus, plasma membrane, cytoplasm, peroxisomes, mitochondria and ribosomes.

2. Which cell organelle is also known as the 'suicide bag'?

- (a) Mitochondria (b) Lysosome
(c) Endoplasmic reticulum (d) Golgi apparatus

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Ans. (b)

Lysosomes are membrane-enclosed organelles that contain an array of enzymes capable of breaking down all types of biological polymers. Lysosomes function as the digestive system of the cell. Lysosomes are known as suicidal bags because once they become damaged and burst open, their digestive enzymes are released and begin eating away at their own cells.

3. Which directs many activities of the cell?

- (a) Ribosome (b) Nucleus
(c) Cell membrane (d) Cytoplasm

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Ans. (b)

The Nucleus is one of the most obvious part of the cell. It is in the middle of the cell and the Nucleus contains all of the cell's chromosomes which encode the genetic material. Nucleus directs many activities of the cell, so this is an important part of the cell.

4. The powerhouse of cell is -

- (a) Cell wall (b) Mitochondria
(c) Ribosome (d) Nucleus
(e) None of the above/More than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (b)

Animal cells and plant cells both contain cell membrane, cytoplasm, and nucleus. The nucleus has two major functions. It stores the cell's hereditary material or DNA and it coordinates the cell's activities within cells. The mitochondrion (plural mitochondria) is a membrane-bound organelle found in most eukaryotic cells. Mitochondria range from 0.5 to 1.0 μ in diameter. These structures are sometimes described as 'The powerhouse of the cell' as they generate energy through respiration. During cellular respiration, the initial process takes place in cytoplasm and then in mitochondria.

5. The substance found in more than 80% of the cell is :

- (a) Protein (c) Mineral
(b) fat (d) water

39th B.P.S.C. (Pre) 1994

Ans. (d)

The amount of water is highest in the cell. Its volume is about 80%.

6. In which cell division, the chromosome number has reduced to half in daughter cell?

- (a) Mitosis (b) Meiosis
(c) Amitosis (d) None of the above

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Ans. (b)

Meiosis is a process where a single cell divides twice to produce four cells containing half the original amount of genetic information. During meiosis one cell divides twice to form four daughter cells. These four daughter cells only have half the number of chromosomes of the parent cell-they are haploid.